SYNBIO REALITY CHECK

Beyond the Green Promise

A field guide to the real-world trade-offs behind synthetic biology

IIT Delhi × IIT Bombay · iGEM collaboration

Two conversations. Six pressure points. One habit: test the claim.

2026 · Discussion booklet



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A route from promise to practice

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Foreword

Hi reader, welcome to the Reality Check.

Synthetic biology can help us make useful products in new ways. But a successful lab result is only one part of a much bigger system. Manufacturing needs energy. Biological feedstocks still use land and water. Engineered strains can change over generations. Products need to be recovered and purified. Labs need reliable supplies, and development needs appropriate oversight.

In SynBio Reality Check, the iGEM teams of IIT Delhi and IIT Bombay put these practical questions at the centre of two conversations. The first looks at manufacturing and environmental trade-offs through IIT Bombay's BiFROST data-centre coolant concept. The second turns to the wider conditions around synthetic biology: supply-chain dependence, variable waste streams, regulation and biosecurity.

Why a reality check?

The aim is not to dismiss the potential of synthetic biology. It is to make our claims more precise. Compare a product with a real alternative. Count the steps beyond the lab. Separate measured results from possible benefits. Be clear about what remains unknown.

A better project is not one with no unanswered questions. It is one that knows which questions still need evidence.

This booklet is an edited guide to the discussions, not a verbatim transcript or a claim that either project has demonstrated commercial-scale performance.

Six pressure points every team should check

01 · Energy

Fermentation may require aeration, mixing, sterilisation and cooling. The electricity source changes the footprint.

2.FEEDSTOCK

A biological input can still have land-use, water and sourcing impacts.

03 · Stability

A production pathway that burdens a cell may lose out to faster-growing variants over repeated generations.

04 · Recovery

Making a molecule is not the same as separating, purifying and preparing it for use.

05 · Access

Reagents, customs delays, cold-chain needs and reliance on a few suppliers can hold up research.

06 · Oversight

Safety controls should reflect the activity, organism, intended use and potential consequences.

The recurring question

What changes when a promising result has to work as part of a real system?

Episode 1 · The hidden trade-offs of manufacturing

Focus	Environmental footprint, feedstocks, strain stability and downstream processing
Project anchor	BiFROST — IIT Bombay's yeast-based data-centre coolant concept
Host	Amogh Kukreja (IIT Delhi)
Respondents	Pratik Ingle and Abhinav Gupta (IIT Bombay)

The first conversation tests the assumption that a bio-based product must automatically be greener. The team discusses a life-cycle assessment (LCA) of its coolant manufacturing route, the energy needed by industrial bioprocesses, the trade-offs around biological feedstocks, and why engineered production can be difficult to sustain over time.

It also looks beyond fermentation to downstream processing—the separation and treatment needed to turn a biological output into a usable product.

THE CENTRAL LESSON

A greener claim must be demonstrated across the relevant stages of production and use, not inferred from the fact that biology is involved.

Watch Episode 1: The hidden trade-offs of manufacturing

Q&A · Does bio-based mean lower carbon?

Q. If a product is made using microorganisms, is it automatically lower-carbon?
A. No. The manufacturing process matters. Fermentation can need substantial heat and electricity for aeration, agitation, sterilisation and cooling. The IIT Bombay team described using the ecoinvent database in an LCA of its coolant production route and noted that India's coal-heavy electricity mix can raise production-stage emissions.

Q. What should a fair comparison include?
A. It should include the inputs and energy needed to manufacture the product, the downstream steps needed to recover it, and relevant impacts during use. The comparison should use a defined conventional alternative and consistent system boundaries.

Q. Could BiFROST still offer a benefit during use?
A. Potentially. The team discussed a cyclic cooling loop: if the coolant improves thermal performance, the energy needed by pumps or chillers could fall over its lifetime. That is a benefit to test and quantify, not one to assume.

REALITY CHECK

“Bio-based” describes how something is made. It does not, by itself, prove a lower life-cycle footprint.

Q&A · Are biological feedstocks always sustainable?

Q. Can switching from petroleum to glucose, cornstarch or sugarcane molasses create new problems?
A. Yes. These inputs can compete with other uses of agricultural land and water. Depending on how they are sourced and how production expands, land-use change and deforestation may also matter.

Q. Does that mean a bio-based product has no advantage?
A. Not necessarily. A bio-based route may reduce dependence on petroleum-derived inputs and exposure to oil-market or geopolitical shocks. But that advantage does not automatically cancel out the footprint of the feedstock or manufacturing process.

Q. What is the honest way to make the claim?
A. Compare against a real alternative, measure the relevant supply chain, and describe the result as a reduction only if the evidence supports it. Do not call a product carbon-neutral without evidence across the system being claimed.

OPEN QUESTION

The discussion identifies land-use and water impacts as important factors, but it does not provide quantified values for them. Those impacts remain items to measure, not benefits or costs to presume.

Q&A · Can an engineered strain keep performing at scale?

Q. Why might a strain that performs well in a short lab run lose productivity later?

A. Producing a target compound can use energy and cellular resources. If the engineered pathway slows growth, variants that produce less may gain a reproductive advantage. A good initial result therefore does not guarantee stable output over many generations.

Q. What did the IIT Bombay team identify as useful checks?

A. Keep metabolic burden in mind during design, optimise fermentation conditions, and test the strain over multiple generations. The team also discussed Flux Balance Analysis (FBA) as a dry-lab tool to examine metabolic constraints and identify pathways that may be affected by an edit.

Q. Can a genetic safeguard guarantee long-term stability?

A. No single safeguard should be treated as infallible. Production stability, containment and performance under intended operating conditions need to be evaluated directly.

REALITY CHECK

Scale-up is not simply a bigger vessel. It also means more time for biological variation and selection to affect performance.

Q&A · What happens after fermentation?

Q. If a microbe makes the target molecule, why is the job not finished?
A. The product may still need to be separated, purified and converted into a usable form. These downstream-processing (DSP) steps can add cost, energy use, heat and solvent requirements—and can materially change the feasibility of the whole route.

Q. How large is the cost?
A. The interview raised 60–80% of production cost as a possible range for DSP. That figure should not be treated as universal: it varies by product and process, and the final booklet should attach a source specific to the claim or leave the number out.

Q. What did the BiFROST team say it was exploring?
A. The team described mapping the post-processing route and exploring in-situ methylation as a possible way to reduce downstream processing. This was described as work to optimise, not as a demonstrated cost or footprint reduction.

REALITY CHECK

A high biological yield is only one measure of success. Recovery, purity, energy and total process cost matter too.

Episode 2 · Access, waste and regulation

Focus	Research supply chains, waste-stream variability, biosafety and biosecurity
Project introduced	DermaSense — IIT Delhi's biologically controlled, air-pollution-responsive moisturiser concept
Host	Abhinav Gupta (IIT Bombay)
Respondent	Amogh Kukreja (IIT Delhi)

The second conversation introduces DermaSense, then widens the lens to challenges across synthetic biology. It asks why labs rely on a small number of suppliers, why real waste is harder to use than a defined laboratory input, and how regulation can support research without overlooking genuine risks.

The discussion is about field-wide constraints; it does not present validation data for DermaSense's product performance.

THE CENTRAL LESSON

Aim for resilient systems and predictable, risk-proportionate oversight—not complete self-sufficiency or deregulation.

Watch Episode 2: Access, waste and regulation

Q&A · How can labs reduce supply-chain dependence?

Q. Why can routine supplies become a research bottleneck?
A. Labs may depend on a small number of overseas suppliers for enzymes, competent cells and synthesis materials. Import costs, distributor mark-ups, customs delays and cold-chain requirements can make timelines and budgets unpredictable.

Q. Does resilience mean every lab should manufacture everything itself?
A. No. The interview argues that complete self-sufficiency is unrealistic. A more practical goal is resilience: multiple suppliers for critical items, plus fallback options where local production is feasible.

Q. What approaches were discussed?
A. The speaker suggested:
• Dual sourcing and targeted in-house fallbacks for routine reagents where quality can be assured.
• Regional reagent foundries that supply reliable, standardised materials to several labs.
• Lyophilisation (freeze-drying) of suitable reagents to reduce dependence on refrigerated transport.
• Workflow redesign, including PCR or assembly approaches that may reduce reliance on long, expensive synthetic DNA constructs.

REALITY CHECK

A fallback is useful only if its quality, cost and readiness have been tested before a shortage occurs.

Q&A · Can real waste become a reliable feedstock?

Q. Why is waste valorisation harder outside the lab?
A. Real agricultural and industrial waste can vary in composition and may contain contaminants or inhibitors. A process optimised for a defined laboratory medium may behave differently when the input changes from batch to batch.

Q. What did the discussion suggest?
A. Standardise or characterise the input wherever possible; build and test tolerance to realistic variation; and design for conditions such as lower pH, imperfect nutrient balance or non-sterile inputs when relevant to the intended process. Adaptive evolution was discussed as one possible way to improve robustness over generations—not a universal fix.

Q. Why does location matter?
A. Dispersed waste is difficult and costly to collect. The speaker recommended looking at concentrated point sources, where a relatively consistent stream is available at one location, rather than assuming scattered waste can be captured efficiently.

REALITY CHECK

"Waste" is not a single feedstock. Start with a specific stream, characterise it, and design for the conditions it actually presents.

Q&A · How can regulation protect safety and support innovation?

Q. How can oversight avoid slowing down every project in the same way?
A. The speaker argued for risk-proportionate pathways. Clearly defined, contained work with well-characterised, low-risk systems may be suitable for simpler procedures, while work involving human use or environmental release warrants stronger scrutiny. This is a policy position from the discussion, not permission to bypass applicable rules.

Q. What else could improve predictability?
A. The interview raised reviewer capacity and clearer pathways for established platforms as areas to improve. Familiarity with a platform may inform a review, but it does not remove the need to assess the actual application and its risks.

Q. Where can biosecurity checks have leverage?
A. The speaker pointed to DNA synthesis providers as an important checkpoint, including sequence screening and customer verification. Such checks can complement—not replace—appropriate oversight elsewhere in the research and deployment process.

REALITY CHECK

Good governance is neither a blanket barrier nor a blanket exemption. It matches safeguards to risk and consequence.

Two episodes. One systems question.

	Episode 1 · Manufacturing	Episode 2 · Access and governance
Main question	Does the product's overall footprint justify the claim?	Can the project get reliable inputs and move responsibly toward use?
Pressure points	Energy, feedstock sourcing, evolutionary stability, downstream processing	Imported reagents, variable waste, regulatory predictability, biosecurity
Useful tools	Life-cycle assessment, process mapping, multi-generation testing, metabolic modelling	Supplier alternatives, reagent quality systems, waste characterisation, risk-based review
Avoid	Calling a product sustainable based only on its biological origin or lab yield	Confusing resilience with complete self-sufficiency, or faster review with weaker safety

What connects them?

A biological design does not exist in isolation. It depends on energy, inputs, facilities, people, processes and rules. Teams need to examine those links early enough to change the design—not only after the technical work is finished.

BiFROST anchors the manufacturing discussion. DermaSense is introduced in the second episode, while the detailed exchange focuses on broader sector-wide constraints rather than reporting product validation.

The iGEM reality-check checklist

Use these questions before writing a project claim, preparing for judging or planning scale-up.

Sustainability

- What is the comparator, and are we comparing the same functional outcome?
- Does our assessment include electricity, feedstocks, downstream processing and performance during use?
- Which land-use, water or emissions impacts are measured—and which remain unknown?

Scale and cost

- Have we tested production stability across multiple generations?
- Have we mapped a plausible recovery and purification route, not just production in the lab?
- Have we identified the costs and conditions that could change the result at scale?

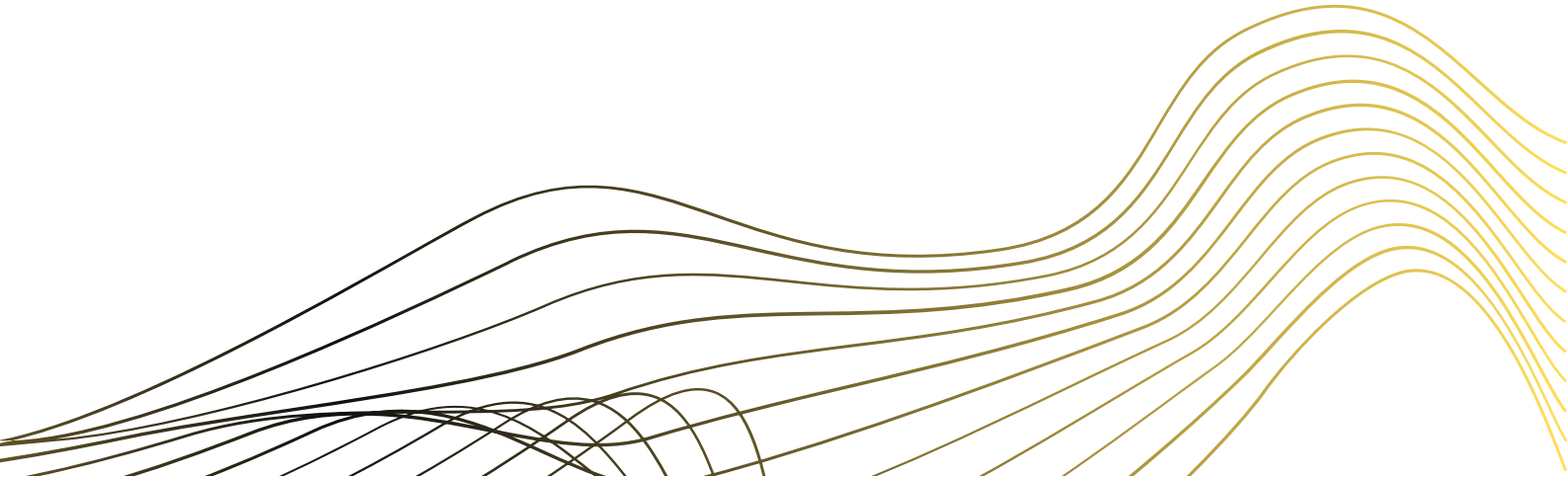
Inputs and waste

- Do we have dependable suppliers or a tested fallback for critical materials?
- Have we characterised the actual waste stream and tested realistic variation?

Safety and governance

- Have we identified the applicable biosafety and regulatory requirements for this specific activity?
- Are safety and biosecurity measures proportionate to the activity and its potential consequences?

The goal is not to eliminate uncertainty. It is to make uncertainty visible—and decide how to test it.



Credibility comes from confronting trade-offs

Synthetic biology does not earn credibility by promising perfection. It earns credibility by making fair comparisons, accounting for the stages that are easy to overlook, designing for biological variability, and building systems that can withstand real-world constraints.

The point of a reality check is not to make ambition smaller. It is to make the ambition testable.

Watch the conversations

Episode 1 — The hidden trade-offs of manufacturing

Episode 2 — Access, waste and regulation

iGEM IIT Bombay wiki: [Insert final page link]

Sources and acknowledgements

• Supplied episode transcripts for both conversations.
• SynBio Reality Check: IIT Delhi × IIT Bombay iGEM Discussion Booklet (planning/content document).
• Is SynBio Really Sustainable: IIT Delhi × IIT Bombay iGEM (outreach write-up draft).
• ecoinvent database used in the team's LCA: add the exact version, datasets and access date.
• Add the original technical references supporting scientific claims, including any source used for the 60–80% DSP range.
• Add approved team credits and photos.